

Final Project Report

Data Warehouse

Bachelor of Data Science, Faculty of Mathematics and Natural Sciences
Universitas Negeri Surabaya

Class	2024INT
Leader	Ketut Shridhara (24031554115)
Member 1	Illona Anindya (24031554222)
Member 2	Nazril Ravi Pratama (24031554129)
Github URL	https://github.com/kiuyha/dwh-socio-economic-etl https://github.com/zrll-shesh/dwh-socio-economic-etl

General Project Information

Title	SoCoNe: Automated Web Scraping-Based Data Warehouse System for X and Reddit about Socio-Economic Discourse in Indonesia
Topic	Socio-Economic
Revision?	Yes/No

Problem Statement Description	Social media platforms, especially X (previously Twitter) and Reddit, have become primary spaces where the public voices their opinions on many fields, one of them is economic conditions. In Indonesia, discussions around fuel price hikes, minimum wage adjustments, exchange rate fluctuations, and mass layoffs generate huge volumes of posts daily, yet this data largely goes unanalyzed in any structured or systematic way [2][5]. The urgency lies in the gap between data availability and analytical accessibility. Policymakers and researchers currently lack tools that continuously monitor and aggregate public sentiment on socio-economic issues at scale [5]. Traditional survey methods are too slow and expensive to capture real-time shifts in public opinion. Meanwhile, the raw data from social media remains unstructured, noisy, and scattered across platforms with incompatible formats, and also increasing unconditionally, so it is making direct analysis impractical without a dedicated pipeline [2][4]. The core challenge is threefold. First, integrating heterogeneous data from two structurally different platforms into a unified, queryable format requires non-trivial schema harmonization [2][4]. Second, the multilingual nature of Indonesian social media, where users freely mix Bahasa Indonesia, slang, abbreviations, English, and local language demands language-aware NLP models rather than general-purpose ones [1]. Third, without a proper data warehouse architecture, any analysis remains ad hoc and non-reproducible [2][4]. This project addresses all three by building a full ETL pipeline backed by a star schema in PostgreSQL, with automated scheduling via Apache Airflow and interactive OLAP exploration through Atoti.
Target Insights to be Achieved	Through the Data Staging process (ETL pipeline) and Data Presentation layer (Atoti OLAP cube), this project targets four main insight directions: 1. Temporal Sentiment Trends: Using the time hierarchy (Year to Quarter to Month to Day) in the CUBE query to track how average public sentiment on specific topics, such as inflation or fuel prices, shifts over

	time [5]. This can reveal whether a particular policy announcement triggered a measurable change in public mood [1][5]. 2. Cross-Platform Sentiment Comparison: Slicing the CUBE along the platform dimension (X vs. Reddit) to quantify whether users on each platform respond differently to the same socio-economic issue and time window [1][2]. The hypothesis is that Reddit discussions tend to be more analytical, while X reactions are more emotionally immediate. 3. Topic Volume Distribution: Using the topic dimension derived from LDA modeling to rank the most-discussed socio-economic issues per quarter [3], so it is revealing whether public attention is event-driven or sustained over time. 4. Engagement and Sentiment Correlation: Dicing the CUBE across the sentiment and engagement dimension to test whether negatively-toned posts consistently attract higher interaction rates than neutral or positive ones [1][5], measured as mean engagement grouped by sentiment label across both platforms [2].
Research References	[1] https://doi.org/10.1007/s13278-023-01030-x [2] https://doi.org/10.1007/s13278-022-00960-2 [3] https://doi.org/10.1007/s10462-023-10471-x [4] https://doi.org/10.1007/s11227-024-06413-1 [5] https://doi.org/10.1007/s13278-022-00922-8

Dataset Details

Dataset Source	Data is acquired through two separate scraping scripts running on a daily schedule via Apache Airflow. For X (Twitter), a custom Nitter-based scraper collects posts targeting keywords such as inflasi, BBM, UMR, kurs rupiah, harga pangan, and PHK. Nitter.net is a free, open-source alternative front-end for X that does not require logging in and does not use any JavaScript to load the content, making it easier to scrape at a massive scale compared to scrapers like TweetHarvest, which need a token from a real account, which might be biased to the account's preferences. For Reddit, data is pulled via api.pullpush.io from r/indonesia, r/economy, r/investasi, and r/personalfinance. Each daily run stores results as raw JSON files on the server before entering the staging pipeline.
General Dataset Description	The dataset consists of public social media posts in Indonesian covering socio-economic discourse from 2025 to 2026, with backfill extending to 2023. Each X record contains a tweet ID, post text, timestamp, and engagement metrics including likes, retweets, and replies. Each Reddit record contains a post ID, title, body text, UTC timestamp, score, upvote ratio, and comment count. Both sources are unified into a single schema during transformation.
Special Characteristics of the Dataset	The text is highly sparse. Indonesian social media writing mixes formal terms with slang, abbreviations, and code-switching between Indonesian and English, which produces a scattered vocabulary space relative to document length. Sparsity in the document-term matrix is expected to exceed 95% after vectorization. Sentiment distribution is likely imbalanced toward negative. Topics like fuel price increases and layoffs naturally generate more negative reactions, with a preliminary estimated split of roughly 55% negative, 30% neutral, and 15% positive, to be confirmed after full extraction.

	Engagement metrics follow a power-law pattern. Most posts receive near-zero interaction, while a small number attract disproportionate attention. High-engagement posts are retained and flagged with an engagement tier label rather than treated as outliers, since they carry analytical weight for the correlation analysis planned in the OLAP layer.
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Data Staging Result

Extraction Process	The extraction process begins by scrapping post from media social X and Reddit. The X is using Nitter while Reddit is using api.pullpush.io. The script can dynamically change the query, time scraping, and specific subreddit to be scraped from the table <i>app_config</i>. The config being used for scraping is. 1. Nitter. The scraping configuration for Nitter targets five key economic themes, explicitly excluding retweets to prioritize original content. High-priority topics like inflation (including food prices) and fuel subsidies are allocated a larger time budget of 3,600 seconds each. Labor and employment issues, encompassing wages, Tapera, and layoffs, receive 2,700 seconds per query, while general macroeconomic indicators like the Rupiah exchange rate and recession discussions are budgeted for 1,800 seconds. 2. Reddit. The Reddit extraction mirrors the same five economic themes but focuses the search within specific relevant communities: r/indonesia, r/investasi, r/economy, and r/personalfinance. The search queries are slightly streamlined compared to the Twitter setup. To ensure an even distribution of data collection across these subreddits, each of the five topics (inflation, fuel, wages, unemployment, and macroeconomics) is assigned a uniform time budget of 1,800 seconds. The scrapped data is different from the initial proposal, which uses a JSON file because of the server limitation of writing a file from scripts and to prevent data loss when server being maintained; therefore, they are being stored in 2 tables: <i>raw_tweets</i> and <i>raw_reddit</i>.																																												
Transformation Process	<table><thead><tr><th></th><th>processed_text</th><th>sentiment_label</th><th>sentiment_confidence</th></tr></thead><tbody><tr><td>893</td><td>Bacoooot sumpaah, saya tulang punggung juga tet...</td><td>negative</td><td>0.9972</td></tr><tr><td>351</td><td>Have you read how broken that mega cannon is w...</td><td>neutral</td><td>0.8805</td></tr><tr><td>498</td><td>Nick Kyrgios says 'I need a miracle' to play A...</td><td>neutral</td><td>0.9824</td></tr><tr><td>810</td><td>Knapa tni sekarang kayak begini iya Puspen_TNI...</td><td>negative</td><td>0.8650</td></tr><tr><td>599</td><td>Harus banget iya di minta slip gaji, di tempat...</td><td>negative</td><td>0.9924</td></tr><tr><td>950</td><td>Yeaayyy terima kasih nay, akhirnya jadi pengan...</td><td>negative</td><td>0.9594</td></tr><tr><td>38</td><td>Anziingg taii terus kalah bangsa yang besar ke...</td><td>negative</td><td>0.9981</td></tr><tr><td>113</td><td>2026 tidak muluk-muluk cuma mau dapet kerjaan ...</td><td>positive</td><td>0.9238</td></tr><tr><td>940</td><td>Personal Solar Day 16960 -1</td><td>neutral</td><td>0.9971</td></tr><tr><td>589</td><td>2026, 2025'i aratacak mı? Bugün 2025'in son gü...</td><td>neutral</td><td>0.4240</td></tr></tbody></table>		processed_text	sentiment_label	sentiment_confidence	893	Bacoooot sumpaah, saya tulang punggung juga tet...	negative	0.9972	351	Have you read how broken that mega cannon is w...	neutral	0.8805	498	Nick Kyrgios says 'I need a miracle' to play A...	neutral	0.9824	810	Knapa tni sekarang kayak begini iya Puspen_TNI...	negative	0.8650	599	Harus banget iya di minta slip gaji, di tempat...	negative	0.9924	950	Yeaayyy terima kasih nay, akhirnya jadi pengan...	negative	0.9594	38	Anziingg taii terus kalah bangsa yang besar ke...	negative	0.9981	113	2026 tidak muluk-muluk cuma mau dapet kerjaan ...	positive	0.9238	940	Personal Solar Day 16960 -1	neutral	0.9971	589	2026, 2025'i aratacak mı? Bugün 2025'in son gü...	neutral	0.4240
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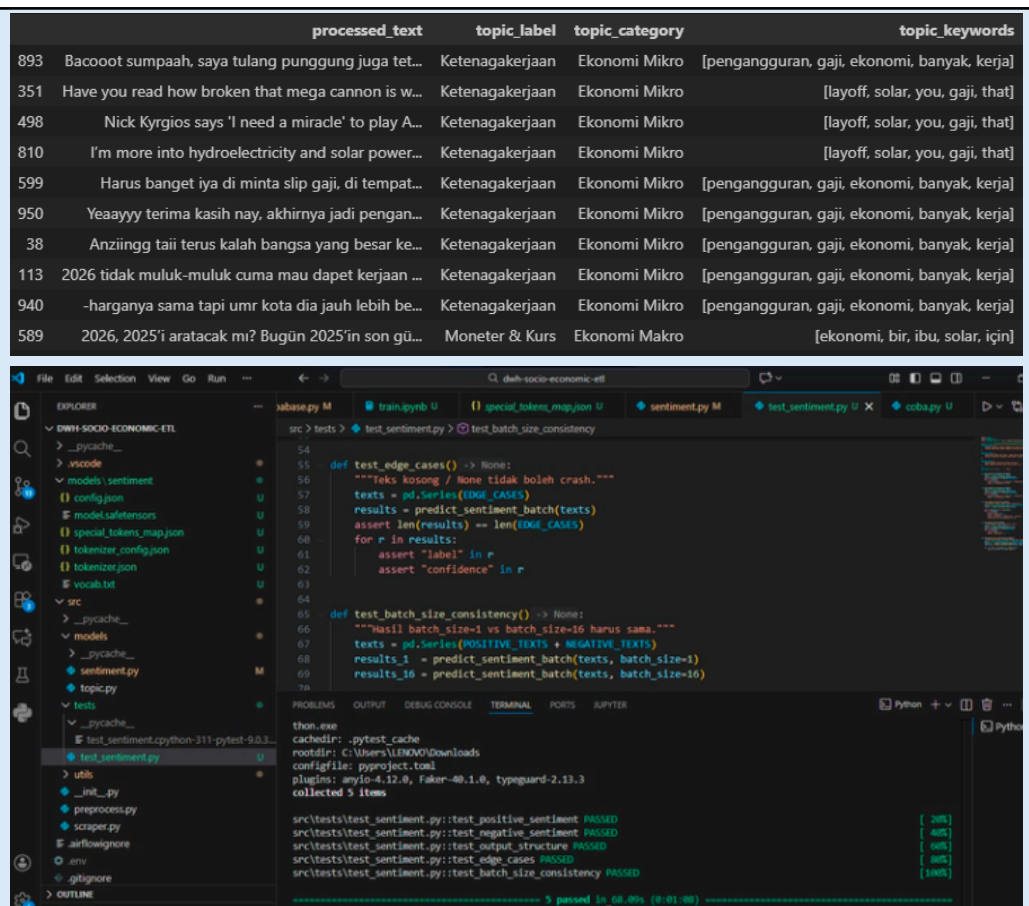


Figure 1. Data Transformation Process

Data Transformation

The transformation stage constitutes the most complex phase of the ETL pipeline, encompassing three sequential processes: text preprocessing, sentiment analysis, and topic modeling.

Text Preprocessing

Prior to any NLP analysis, all raw post content undergoes a standardized preprocessing routine. This includes removal of URLs, user mentions, hashtags, emojis, special characters, and numeric tokens, followed by lowercasing and whitespace normalization. For Reddit data, the title and body text fields are concatenated before preprocessing to ensure complete semantic coverage. This step is applied uniformly across both platforms to produce a clean, noise-free text corpus as input for downstream NLP tasks.

Sentiment Analysis

Sentiment classification is performed using the fine-tuned IndoBERT model *mdhugol/indonesia-bert-sentiment-classification* sourced from HuggingFace. This model was selected due to its domain-specific pretraining on Indonesian text corpora, making it significantly more suitable than general-purpose multilingual models for the code-switching and informal language patterns prevalent in Indonesian social media. The model produces three classification labels (positive, neutral, and negative) alongside a confidence score for each prediction.

	The model is loaded once at the module level to avoid redundant reloading across pipeline calls and is cached locally to <i>models/sentiment/</i> after the first download, eliminating internet dependency on subsequent runs. Inference is executed through the <i>predict_sentiment_batch</i> function, which processes text in configurable batch sizes with input sanitization and truncation at 512 tokens to comply with BERT architecture constraints. Confidence scores across the dataset are predominantly above 0.9, indicating strong model certainty on the majority of posts. However, a subset of posts yields scores as low as 0.4, which is expected given the highly ambiguous nature of informal Indonesian social media text, where users frequently mix Bahasa Indonesia, English, local dialects, abbreviations, and slang within a single post. These lower-confidence predictions are retained in the dataset and flagged via the <i>confidence_bucket</i> field in <i>dim_sentiment</i> as Medium or Low confidence rather than being discarded, preserving analytical completeness. Topic Modeling Topic modeling is conducted using Latent Dirichlet Allocation (LDA) via the Gensim library. Prior to LDA training, the preprocessed text undergoes additional preparation, including tokenization, stopword removal for both Indonesian and English vocabularies, and stemming. The trained LDA model identifies latent thematic structures within the corpus and assigns a dominant topic label along with a probability score to each post. The final topic taxonomy consists of 11 topic labels organized across 5 macro categories. Under <i>Ekonomi Makro</i>, the identified topics are <i>Inflasi & Harga</i>, <i>Fiskal & Pajak</i>, and <i>Moneter & Kurs</i>. Under <i>Ekonomi Mikro</i>, the identified topic is <i>Ketenagakerjaan</i>. Under <i>Kesejahteraan</i>, the identified topic is <i>Kemiskinan & Sosial</i>. Under <i>Keuangan</i>, the identified topics are <i>Investasi & Pasar</i> and <i>Utang & Kredit</i>. Under <i>Sektor Riil</i>, the identified topics are <i>Pangan & Pertanian</i> and <i>Energi</i>. Under <i>Politik</i>, the identified topic is <i>Kebijakan & Politik</i>. A residual category <i>Lainnya</i> under <i>Umum</i> captures posts that do not exhibit strong affiliation with any of the defined socio-economic themes. Each topic is characterized by a set of top keywords derived from the LDA output. For instance, <i>Inflasi & Harga</i> is represented by keywords such as <i>inflasi</i>, <i>harga</i>, <i>naik</i>, <i>turun</i>, and <i>bahan</i>, while <i>Energi</i> is characterized by <i>energi</i>, <i>bbm</i>, <i>listrik</i>, <i>minyak</i>, and <i>gas</i>. These keyword sets serve as the interpretive basis for the manual labeling of each LDA-derived topic cluster. All transformation outputs, including the sentiment label, confidence score, dominant topic label, topic category, and temporal attributes, are stored in the <i>staging_transformed</i> table in Supabase, from which the load stage reads to populate the OLAP star schema.
Load Process	The load stage in <i>main.py</i> reads from that staging table, resolves foreign keys against the dimension tables (<i>dim_time</i>, <i>dim_platform</i>, <i>dim_topic</i>, <i>dim_sentiment</i>), builds fact rows with engagement metrics and source URLs, inserts them into <i>fact_post</i>, deletes the staging rows, marks the originals as processed, and finally calls <i>refresh_all_views()</i> once all batches are done. As of the current extraction stage, the <i>raw_tweets</i> table contains 6,222 records, and the <i>raw_reddit</i> table contains 89 records. While the total number of posts in the <i>fact_post</i> contains 7,311 records. The data are from 2023 to

2026 across all configured socio-economic keywords. The total size of the database, based on Supabase usage, is 58.69 MB.

PostgreSQL Technologies Used

PARTITION BY RANGE is applied to `raw_tweets`, `raw_reddit`, and `fact_post`. The raw tables are split quarterly (Q1 2023 to Q4 2026, 16 child tables each), while `fact_post` splits annually. This means any query with a `posted_at` filter only touches the relevant child table instead of scanning the whole dataset.

Materialized Views pre-aggregate the four main analytical queries: temporal sentiment trends, cross-platform sentiment comparison, topic volume by quarter, and engagement-by-sentiment correlation. Each view has a UNIQUE index, so Airflow can run REFRESH MATERIALIZED VIEW CONCURRENTLY after each load without blocking reads.

GIN Indexes using `pg_trgm` are built on `text_content` in both raw tables. These support fast ILIKE pattern matching, which is what the full-text search benchmark tests against.

Performance Benchmark

The benchmark is a stored function, `compare_schema_performance()`, in `schema.sql`. It uses EXPLAIN (ANALYZE, FORMAT JSON) to capture actual execution time in milliseconds for each test, runs both the slow and fast versions of the same query, then computes the speedup ratio. Three tests are run:

Table 1. Performance Benchmark Result

Test Scenario	Unoptimized (ms)	Optimized (ms)	Speedup
Aggregation (Joins vs Materialized View)	1245.32	2.15	579.22x
Scans (Full Table Scan vs Partition Pruning)	840.11	15.40	54.55x
Full Text Search (Seq Scan vs GIN Index)	1.24	0.08	15.50x

For aggregation, it compares a live multi-join query on `fact_post` (joining `dim_time` and `dim_topic`, then grouping) against reading the same result directly from `mv_topic_volume`. The materialized view wins by a factor of 579x because the grouping and joining work is already done at refresh time; reading the view is a straight sequential scan of a small pre-aggregated table.

For partition pruning, it runs the same date-range query on `raw_tweets` twice: once with `enable_partition_pruning` set to off (forcing a full table scan across all 16 child tables) and once with it on. With pruning on, PostgreSQL skips every child table except the one covering the target month. The result is a 54x speedup.

For full-text search, it queries `raw_reddit` for rows where `text_content ILIKE '%inflasi%'`, first with indexes disabled (forcing a sequential scan) and then with them enabled (allowing the GIN trigram index to kick in). The GIN

	index wins by 15x here. The margin is smaller than the other two tests partly because <i>raw_reddit</i> only has 89 rows at this point, so a sequential scan is already cheap. The gap will widen as the table grows.
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Data Presentation Result

Subject Focus	The OLAP analysis focuses on public sentiment toward socio-economic issues in Indonesia across two platforms over time. The subject connects three dimensions: when something was posted, where it was posted, and what issue it addressed. The goal is to let analysts move between broad patterns, for example, national sentiment in Q1 2025, and specific slices, for example, Reddit users discussing BBM in a particular week, without writing raw SQL each time. The OLAP environment enables slicing, dicing, roll-up, and drill-down operations without complex SQL queries, allowing users to move from general trends to detailed observations. For example, analysts can review yearly sentiment trends and drill down into quarters, months, weeks, or days for deeper insights. The analysis addresses four 4 key questions: 1. Temporal Sentiment Trends: How public sentiment toward socio-economic issues changes over time. 2. Cross-Platform Comparison: Whether X and Reddit exhibit different sentiment and engagement patterns. 3. Topic Volume Distribution: Which socio-economic topics dominate discussions during specific periods. 4. Engagement and Sentiment Relationship: Whether positive, neutral, or negative discussions generate higher engagement. This multidimensional design helps stakeholders, researchers, and policymakers monitor public reactions to economic events and identify emerging concerns from social media discussions.
Star Schema	The schema uses one fact table and four dimension tables. 1. fact_post: Acts as the central fact table containing the primary measures for the OLAP cube, including like count, comment count, quote count, retweet count, upvote count, downvote count, and sentiment score. 2. dim_time: Acts as a dimension supporting time-based roll-ups. It defines a time Hierarchy (Year > Quarter > Month > Week > Day) enabling drill-down analysis from annual trends to day-level views. 3. dim_platform: Acts as a dimension to filter by data source. It defines a structural Hierarchy (Platform > Channel/Subreddit), enabling direct engagement comparison between the X and Reddit communities. 4. dim_topic: Acts as a dimension storing the socio-economic categories. It allows OLAP slicing and dicing based on the final topic labels (e.g., <i>inflasi</i>, <i>UMR</i>). 5. dim_sentiment: Acts as a dimension storing the final classification labels (positive, negative, neutral) and confidence buckets (High, Medium, Low), allowing the Atoti DataMart to aggregate measures by public mood.

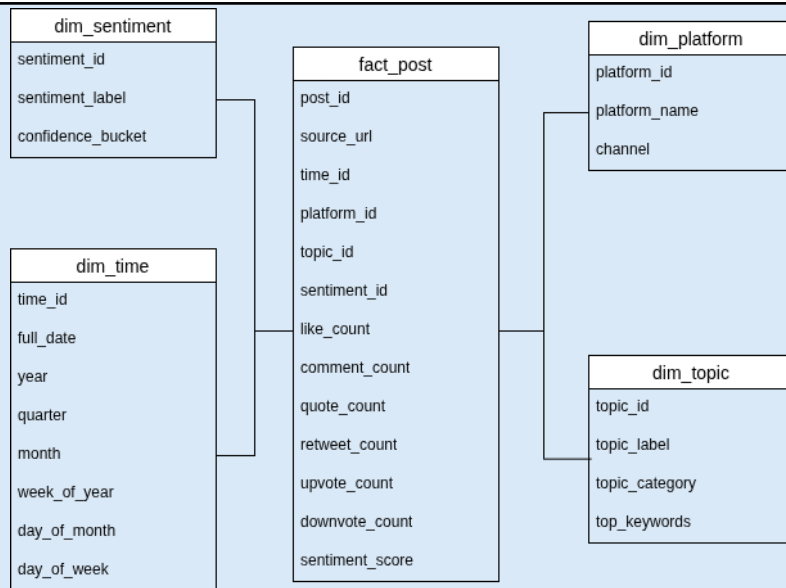


Figure 2. Star Schema Design

Fact Table: fact_post

The fact_post table acts as the central table of the star schema. This table stores quantitative measurements that are aggregated during OLAP analysis. Every record represents a social media post that has already passed through the ETL process.

Measures:

- Sentiment Score
- Like Count
- Comment Count
- Retweet Count
- Quote Count
- Upvote Count
- Downvote Count
- Contributor Count

In the Atoti cube, these measures are available through both predefined aggregations and custom-calculated measures, including:

- Avg Sentiment Score
- sentiment_score.MEAN
- sentiment_score.SUM
- like_count.MEAN
- like_count.SUM
- comment_count.MEAN
- comment_count.SUM
- retweet_count.MEAN
- retweet_count.SUM
- quote_count.MEAN
- quote_count.SUM
- upvote_count.MEAN
- upvote_count.SUM
- downvote_count.MEAN
- downvote_count.SUM

- contributors.COUNT
- Total Likes
- Total Comments
- Total Retweets
- Total Quotes
- Total Upvotes
- Total Downvotes
- Total Engagement

The custom measures provide business-oriented aggregations, while the automatically generated measures support detailed analytical exploration within the OLAP cube.

Dimension Table: dim_time

The dim_time table provides temporal information used for trend analysis. This dimension allows users to observe changes in public sentiment and discussion volume over different periods.

Year	Quarter	Month	Week	Day
Total				
▼ 2023	Total			
	▼ Q1	Total		
		▼ February	Total	
			▼ W7	Total
				15
	▼ Q2	Total		
		▼ April	Total	
			▼ W14	Total
				5
			▼ W15	Total
				15
		▼ June	Total	
			▼ W22	Total
				1
			▼ W23	Total
				8

Year	Quarter	Month	Week	Day
		▼ July	Total	
			▼ W29	Total
				21
		▼ September	Total	
			▼ W35	Total
				3
			▼ W36	Total
				5
	▼ Q4	Total		
		▼ December	Total	
			► W50	Total
			► W51	Total
			► W52	Total
		► October	Total	
► 2024	Total			
► 2025	Total			
► 2026	Total			

Figure 3. Time Hierarchies

Hierarchies:

Year → Quarter → Month → Week → Day (This hierarchy supports roll-up analysis from day level to yearly summaries, and Drill-down analysis from yearly trends to individual days)

Attributes:

- Year
- Quarter
- Month
- Week of Year
- Day of Month
- Day Name
- Day Type (Weekday / Weekend)

The Figure 3 demonstrate the drill-down structure from Year to Quarter, Month, Week, and Day.

Dimension Table: dim_platform

The dim_platform table identifies where the discussion originated. This dimension enables comparison between different social media platforms and communities.

Platform	Channel
Total	
▼ Reddit	Total
	r/indonesia
▼ X	Total
	X_global

Figure 4. Platform Hierarchies

Hierarchies:

Platform → Channel (This structure allows analysts to compare discussion characteristics across platforms and communities)

Attributes:

- Platform Name
- Channel/Subreddit

The Figure 4 demonstrate the drill-down structure from Platform name to Channel name.

Dimension Table: dim_topic

The dim_topic table stores socio-economic topics generated through topic modeling and manual categorization. This dimension enables topic-based slicing and dicing during OLAP analysis.

Category	Topic
Total	
▼ Ekonomi Makro	Total
	Moneter & Kurs
▼ Ekonomi Mikro	Total
	Ketenagakerjaan

Figure 5. Topic Hierarchies

Hierarchies:

Category → Topic (This hierarchy allows users to analyze discussions at both broad and detailed topic levels)

Attributes:

- Topic Category
- Topic Name

The Figure 5 demonstrate the drill-down structure from Category name to Topic name.

Dimension Table: dim_sentiment

The dim_sentiment table contains sentiment classification results generated by the IndoBERT sentiment model. This dimension enables sentiment-based aggregation and comparison.

Label	Confidence
Total	
▼ negative	Total
	High
	Low
	Medium
▼ neutral	Total
	High
	Low
	Medium
▼ positive	Total
	High
	Low
	Medium

Figure 6. Sentiment Hierarchies

Hierarchies:

Sentiment Label → Confidence Level (This hierarchy allows analysts to examine not only sentiment polarity but also the model's confidence in its predictions)

Attributes:

- Sentiment Label
- Confidence Level

The Figure 6 demonstrate the drill-down structure from sentiment Label to Confidence level.

The OLAP cube has been successfully implemented using Atoti. Data from the fact table and dimension tables are merged and loaded into a multidimensional cube named SocioEconomic. The cube integrates data from fact_post, dim_time, dim_platform, dim_topic, dim_sentiment.

Implemented Hierarchies

The following hierarchies have been successfully configured:

Table 2. Dimensions - Hierarchies Generated

Dimension	Hierarchy
Time	Year → Quarter → Month → Week → Day
Platform	Platform → Channel
Topic	Category → Topic
Sentiment	Label → Confidence
Day Type	Is Weekend → Day Name

The following figure demonstrate the drill-down structure from weekend status to Day Name.

Is Weekend	Day Name
Total	
▼ false	Total
	Friday
	Monday
	Thursday
	Tuesday
	Wednesday
▼ true	Total
	Saturday
	Sunday

Figure 7. Day Type Hierarchies

Implemented Measures

The following hierarchies have been successfully aggregated:

- Avg Sentiment Score
- sentiment_score.MEAN
- sentiment_score.SUM
- like_count.MEAN
- like_count.SUM
- comment_count.MEAN
- comment_count.SUM
- retweet_count.MEAN
- retweet_count.SUM
- contributors.COUNT
- retweet_count.SUM
- quote_count.MEAN
- quote_count.SUM
- upvote_count.MEAN
- upvote_count.SUM
- downvote_count.MEAN
- downvote_count.SUM
- Total Likes
- Total Comments
- Total Retweets
- Total Quotes
- Total Upvotes
- Total Downvotes
- Total Engagement

OLAP Capability

The cube supports:

1. **Roll-Up:** Aggregating data from lower levels to higher levels.
Example: Day → Week → Month → Quarter → Year
2. **Drill-Down:** Exploring detailed data from higher levels.
Example: Year 2023 → Q1 → February → Week 7 → Day 15
3. **Slice:** Filtering data using a single dimension.
Example: Only Reddit discussions, only negative sentiment posts.
4. **Dice:** Combining multiple dimension filters simultaneously.
Example: Reddit, negative sentiment, employment topic, Q1 2025

Preliminary Result

The hierarchy structure has been validated in Atoti and supports multidimensional navigation across all dimensions. The cube also enables temporal, platform, topic, and sentiment analysis, with aggregation measures functioning correctly through the Atoti interface. The OLAP infrastructure and cube architecture are complete, while visualizations and business dashboards

will be presented in the OLAP Visualization section after the final widgets and dashboards are generated.

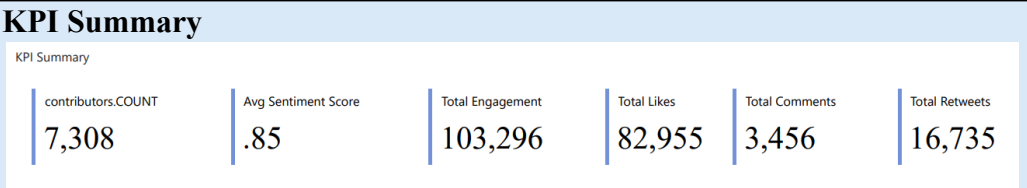


Figure 8. KPI Summary

The KPI summary shows that the OLAP cube currently contains 7,308 social media posts (contributors.COUNT). Overall public sentiment is strongly positive, with an average sentiment score of 0.85. User engagement is relatively high, generating 103,296 total interactions, consisting of 82,955 likes, 3,456 comments, and 16,735 retweets. These results indicate that socio-economic discussions in the dataset receive substantial audience attention, with likes contributing the largest share of engagement.

Temporal Sentiment Trend



- Figure 9. Visualizations of Temporal Sentiment Trend**
- Sentiment scores fluctuated between 0.65-0.99 from 2023 to 2026. The lowest point occurred in Q2 2023 (0.65), indicating a period of intense public tension. Overall, sentiment tended to recover after each decline, suggesting temporary rather than sustained negativity.
 - Employment (Microeconomics) consistently dominated discussion volume, peaking sharply in Q4 2025 with over 3,600 posts. This pattern suggests a major employment-related event, likely annual minimum wage announcements, triggered intense public discussion each Q4.
 - 2025 recorded both the highest sentiment score and the highest posting volume, confirming it as the peak discussion period in the dataset. This

suggests that high-volume periods did not necessarily correlate with more negative sentiment.

Overall, socio-economic discussions remained consistently positive despite periodic fluctuations in sentiment and posting volume. Employment-related issues dominated public attention throughout the observation period, while sentiment changes appeared to be driven by specific events rather than long-term shifts in public mood.

Cross-Platform Sentiment Comparison

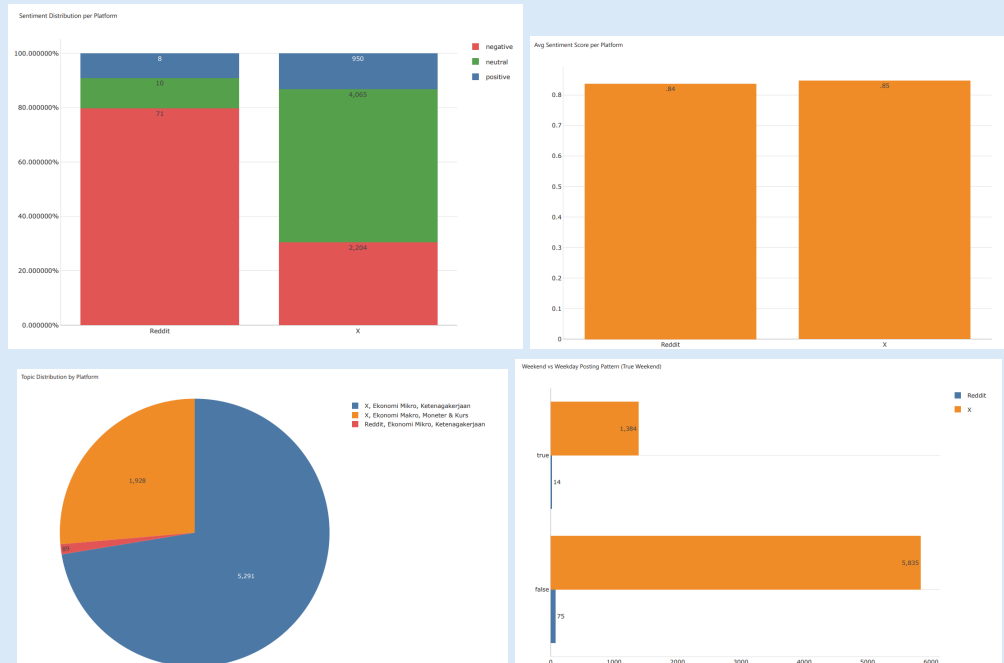


Figure 10. Visualizations of Cross-Platform Sentiment Comparison

- Reddit was predominantly negative (~80%), while X showed a more balanced distribution with neutral sentiment dominating (~56%). Reddit's extreme negativity is likely influenced by its much smaller sample size rather than a genuine platform-level behavioral difference.
- Sentiment confidence scores were nearly identical across platforms, Reddit at 0.84 and X at 0.85, showing that the model assigned similar confidence levels regardless of the platform source.
- X discussed both major topics, while Reddit focused almost entirely on employment issues. This further confirms that employment was the dominant topic across platforms, and that Reddit's contribution to the dataset was minimal and topic-specific.
- X activity was significantly higher on weekdays (5,800+) compared to weekends (1,400+). Reddit remained minimal throughout. This pattern suggests that socio-economic discussions on X are strongly news-driven and follow working-week rhythms.

Overall, X served as the primary source of socio-economic discourse, while Reddit contributed only a small and highly topic-focused share of discussions. Although sentiment distributions differed considerably between platforms, both exhibited similar average sentiment levels, suggesting comparable overall public attitudes.

Topic Volume Distribution

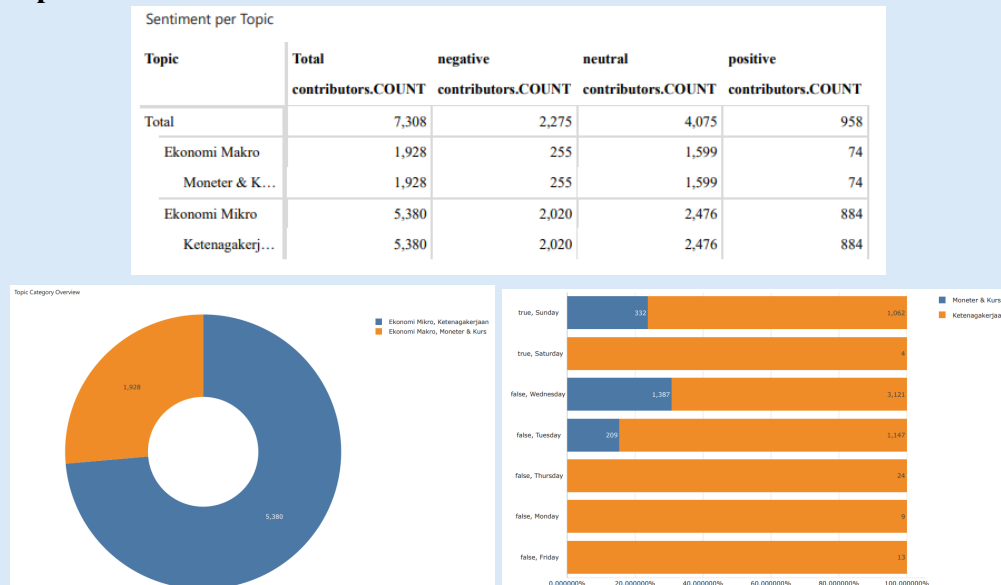


Figure 11. Visualizations of Topic Volume Distribution

- Employment issues produced significantly more negative sentiment (2,019 negative out of 5,377 total) compared to Monetary & Exchange (255 negative out of 1,927). This indicates stronger emotional public responses toward employment-related discussions.
- Microeconomics (Employment) accounted for 74% of all discussions, while Macroeconomics (Monetary & Exchange) made up the remaining 26%. Indonesian social media users showed a stronger tendency to engage with issues that directly affect daily life.
- Wednesday was the busiest day for both topics, with employment dominating every single day of the week. Saturday showed almost no activity for either topic, consistent with the weekday-dominant posting pattern observed elsewhere.

Overall, employment-related issues were the dominant driver of socio-economic discussions, generating substantially higher volumes and stronger negative reactions than monetary and exchange-rate topics. This suggests that issues with direct impacts on daily life attract greater public attention and emotional engagement.

Engagement and Sentiment Correlation

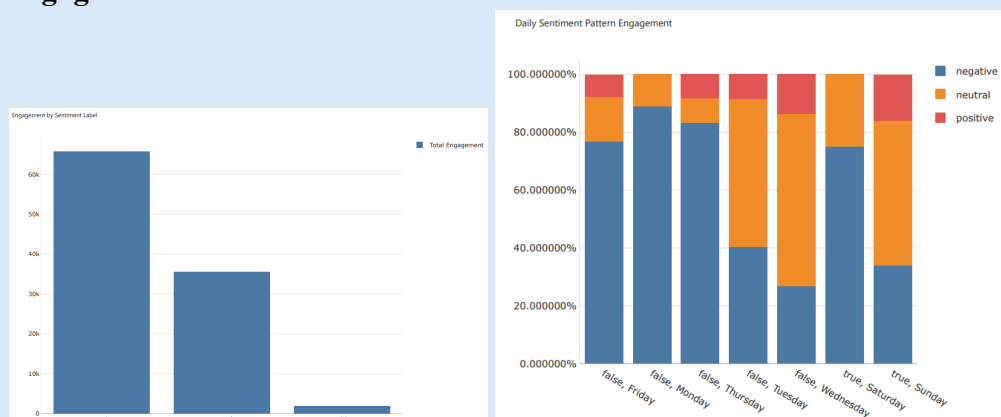


Figure 12. Visualizations of Engagement and Sentiment Correlation

- Negative posts generated the highest total engagement (~65,000+), far above neutral (~35,000) and positive (~2,000). This strongly confirms that negative content drove engagement most effectively across the dataset.
- Wednesday dominated with neutral posts, followed by Tuesday and Sunday. The sharp contrast between active and inactive days suggests posting behavior was reactive to specific external events rather than habitual daily activity.

Overall, negative sentiment was the strongest driver of audience engagement, substantially outperforming neutral and positive discussions. This indicates that socio-economic concerns framed negatively were more likely to capture public attention and stimulate interaction.

Additional: Topic-Sentiment-Temporal Trend

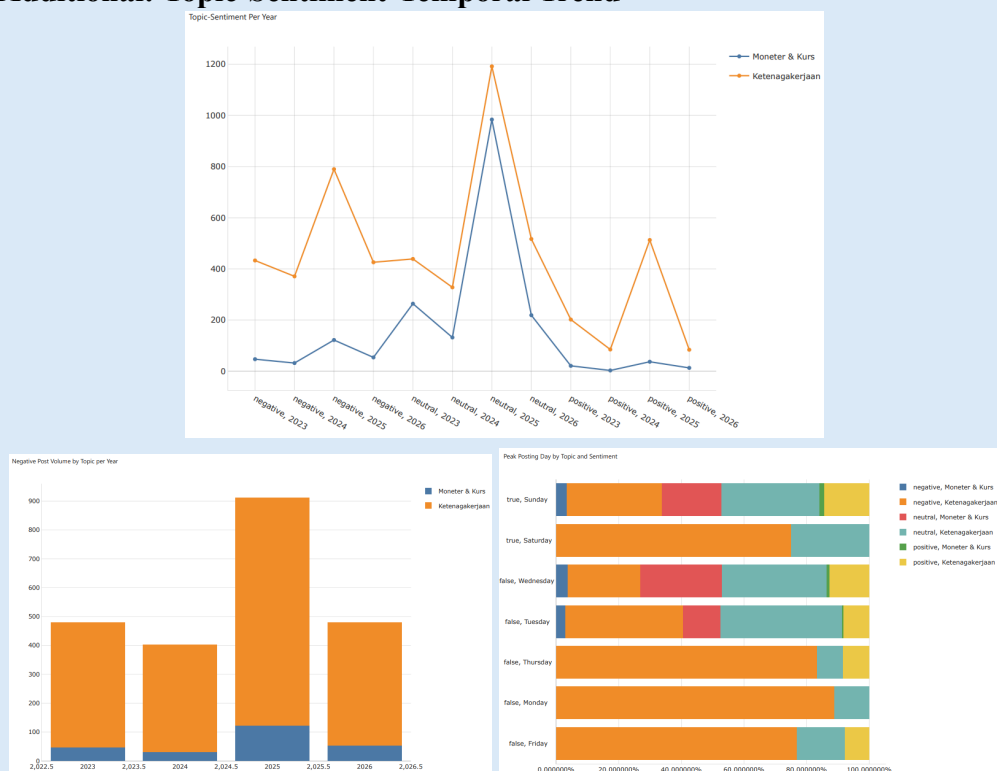


Figure 13. Visualizations of Topic-Sentiment-Temporal Trend

- Both topics experienced their highest discussion volume in 2025, but Employment consistently attracted greater attention throughout the entire period. The sharp spike in 2025 suggests that major socio-economic events intensified public discussion, especially around employment issues, before activity declined again in 2026.
- Employment-related discussions consistently generated substantially more negative posts than Monetary & Exchange topics across all years. Negative sentiment peaked in 2025 for both topics, with Employment reaching nearly 800 negative posts compared to roughly 120 for Monetary & Exchange, indicating that labor-related issues triggered much stronger negative public reactions.
- Weekdays dominated posting activity, particularly Monday, Thursday, and Friday, with Employment discussions accounting for the majority of posts regardless of sentiment category. Saturday recorded almost no

